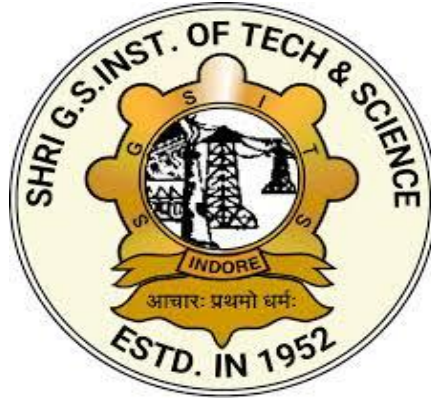


**SHRI GOVINDRAM SEKSARIA INSTITUTE OF
TECHNOLOGY & SCIENCE , INDORE**



DEPARTMENT OF INFORMATION TECHNOLOGY

Minor Project Report

On

**GlucoAI Monitor — AI-Based Non-Invasive Glucose
Monitoring System**

Supervised by:

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**SHRI GOVINDRAM SEKSARIA INSTITUTE OF TECHNOLOGY &
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DEPARTMENT OF INFORMATION TECHNOLOGY



RECOMMENDATION

We are pleased to recommend that the Minor Project Report entitled **“GlucoAI Monitor — AI-Based Non-Invasive Glucose Monitoring System”** submitted by **Dilpreet Singh Gill, Harish Gupta, Hemant Dixit** be accepted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Information Technology of Rajiv Gandhi Pradyogiki Vishwavidyalaya (RGPV), Bhopal (M.P.), during the academic session 2026.

[Prof. K. K. Sharma]

[Assistant Professor]

[Head of Department]

[Professor]

[Information Technology]

[Information Technology]

**SHRI GOVINDRAM SEKSARIA INSTITUTE OF TECHNOLOGY &
SCIENCE , INDORE**

DEPARTMENT OF INFORMATION TECHNOLOGY]



CERTIFICATE

This is to certify that the Minor Project entitled “**GlucoAI Monitor — AI-Based Non-Invasive Glucose Monitoring System**” submitted by [Student’s Name] is a bonafide work carried out under my supervision and is being submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Information Technology of Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal, during the academic session 2026–2027 has been examined and evaluated by us. The report is found to be satisfactory and meets the academic requirements prescribed by the institution.

Internal Examiner
Date:

External Examiner
Date:

DECLARATION

I, *Dilpreet Singh Gill, Harish Gupta, Hemant Dixit*, an undergraduate student in the Department of Information Technology, S.G.S.I.T.S., Indore (M.P.), hereby declare that the Minor Project titled **“GlucoAI Monitor — AI-Based Non-Invasive Glucose Monitoring System”** is my original work, carried out under the supervision of **Sonu Airen, Assistant Professor, Department of Information Technology, S.G.S.I.T.S., Indore (M.P.)**.

I further declare that, to the best of my knowledge, this dissertation does not contain any material previously submitted for the award of any degree or diploma in this or any other university. Any sources of information used have been properly acknowledged and cited.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our guide, **Sonu Airen, Professor**, for his valuable guidance, continuous support, and encouragement throughout the completion of this project.

We are also thankful to the **Head of the Department, Prof. K. K. Sharma**, for providing the necessary facilities and support to carry out this work.

We extend our heartfelt thanks to the **Dean (ARSD), Prof. Lalit Purohit**, for his encouragement and institutional support.

Finally, we are grateful to the faculty members and technical staff of the **Department of Information Technology, S.G.S.I.T.S., Indore**, for their support and cooperation.

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ABSTRACT

This project presents an innovative approach to **non-invasive glucose monitoring** using Artificial Intelligence and Computer Vision techniques. Traditional glucose measurement methods require invasive procedures such as finger-prick tests, which can be uncomfortable, painful, and inconvenient for users — particularly for diabetic patients who must monitor their levels multiple times per day. The proposed system, **GlucoAI Monitor**, eliminates this requirement by analyzing facial/fingertip video data to estimate blood glucose levels through optical sensing.

The system captures a short video of the user's fingertip (placed on the smartphone camera with flash on) and processes it to extract **Photoplethysmographic (PPG) signals**, which represent subtle variations in skin color caused by pulsatile blood flow through peripheral capillaries. These PPG signals encode physiological information about cardiac activity, vascular tone, and blood composition — all of which correlate with blood glucose concentration.

The extracted raw signal undergoes a multi-stage preprocessing pipeline: 1. **Motion artifact removal** using high-pass filtering at 0.3 Hz 2. **Adaptive bandpass filtering** (0.7–3.5 Hz) using Chebyshev Type II filters with automatic frequency adaptation based on spectral analysis 3. **Savitzky-Golay smoothing** for noise reduction while preserving peak morphology 4. **Z-score normalization** for standardization

The processed signal is then analyzed to extract **28 comprehensive features** across six categories: statistical, peak-based, energy, frequency-domain, morphological, and nonlinear features. These features, along with the raw signal waveform, are passed into a **Multi-Scale CNN + Bidirectional LSTM + Self-Attention** deep learning model trained on time-series data to predict glucose levels. The application is implemented using a modern web-based interface built with **React (Vite)** for the frontend and a **Flask** backend for video processing and model inference. The system includes user authentication with SQLite database, real-time PPG waveform visualization, glucose and heart rate estimation with confidence scoring, and prediction history tracking.

The final model achieves a **Mean Absolute Error (MAE) of ~5 mg/dL** and an **R² score of 0.957** on the experimental dataset, demonstrating the potential of combining AI and healthcare technologies to create a non-invasive, user-friendly, and scalable glucose monitoring solution.

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CHAPTER 1

INTRODUCTION

1.1 Background

Diabetes mellitus is one of the most prevalent chronic metabolic disorders affecting millions of people worldwide. According to the International Diabetes Federation (IDF), approximately **537 million** adults (aged 20–79) were living with diabetes globally in 2021, and this number is projected to rise to **783 million by 2045**. In India alone, over **77 million** people are estimated to have diabetes, making it the “diabetes capital of the world.”

Effective management of diabetes requires **frequent monitoring of blood glucose levels** — often multiple times daily for individuals on insulin therapy. The current gold standard for self-monitoring involves **finger-prick capillary blood glucose testing**, where a small lancet punctures the fingertip to draw a blood sample, which is then analyzed by a portable glucometer. While accurate, this method has several significant drawbacks:

- **Pain and discomfort:** Repeated finger pricks cause soreness, calluses, and reduced sensitivity in fingertips over time.
- **Infection risk:** Each puncture creates an open wound, introducing potential infection pathways.
- **Compliance issues:** Due to pain and inconvenience, many patients skip glucose measurements, leading to poor disease management.
- **Consumable costs:** Test strips are expensive, costing patients significant amounts annually.

With rapid advancements in **Artificial Intelligence (AI)**, **Deep Learning**, and **Computer Vision**, it is now possible to explore entirely non-invasive alternatives for glucose estimation. **Photoplethysmography (PPG)** — the optical technique of detecting volumetric changes in blood in peripheral circulation — has emerged as a promising

modality. PPG signals can be captured simply by pressing a finger against a smartphone camera, making the technique accessible to anyone with a mobile device.

This project, **GlucoAI Monitor**, leverages PPG signal analysis combined with advanced deep learning architectures (Multi-Scale CNN + Bidirectional LSTM with Attention) to predict blood glucose levels from short video recordings — completely non-invasively.

1.2 Problem Statement

Existing glucose monitoring systems present several challenges that limit their effectiveness and adoption:

Problem	Description
Invasive & Painful	Conventional finger-prick tests require blood samples, causing pain, skin damage, and reduced compliance
Expensive	Continuous Glucose Monitors (CGMs) cost ₹5,000–₹15,000 per sensor, test strips cost ₹15–₹50 each
Inconvenient	Carrying lancets, test strips, and glucometers; finding hygienic testing conditions
Infection Risk	Each finger prick creates an open wound susceptible to infection
Low Compliance	Due to the above factors, patients often skip measurements, leading to poor diabetes management
Not Scalable	Mass screening programs are impractical with invasive methods

There is a critical need for a **non-invasive, affordable, smartphone-based** glucose monitoring solution that can democratize glucose monitoring and improve patient compliance.

1.3 Objectives

The primary objectives of this project are:

1. **Develop a non-invasive glucose monitoring system** that uses smartphone camera video to estimate blood glucose levels without any blood sample.
2. **Extract PPG (Photoplethysmographic) signals** from fingertip video recordings using computer vision and image processing techniques.
3. **Implement advanced signal processing** including motion artifact removal, adaptive bandpass filtering, and signal quality assessment.

4. **Design and train a deep learning model** (Multi-Scale CNN + Bidirectional LSTM + Self-Attention) to predict glucose levels from extracted PPG features.
5. **Build a full-stack web application** with React frontend and Flask backend providing user authentication, video upload, real-time results, and history tracking.
6. **Provide real-time visualization** of PPG waveforms, glucose levels, heart rate, and confidence scores.
7. **Achieve clinically meaningful accuracy** with $MAE < 15$ mg/dL for experimental predictions.

1.4 Scope

The scope of this project extends across multiple domains and applications:

- **Healthcare Monitoring:** Primary use as a supplementary glucose monitoring tool for diabetic patients between traditional measurements.
- **Preventive Health Screening:** Enabling mass screening for pre-diabetes in resource-limited settings using only a smartphone.
- **Wearable Technology Integration:** The PPG extraction and AI inference pipeline can be adapted for integration with smartwatches and fitness bands that have optical heart rate sensors.
- **Remote Patient Monitoring (RPM):** Enabling telemedicine applications where patients can share glucose estimates with their physicians remotely.
- **Research Platform:** Serving as a research framework for further studies on non-invasive biomarker estimation from optical signals.
- **Mobile Health (mHealth):** Future migration to a native mobile application for iOS and Android platforms.

Disclaimer: This system is an experimental research prototype. It is NOT a certified medical device and should NOT be used for clinical decision-making. Always consult a healthcare professional and use a certified glucometer for accurate readings.

CHAPTER 2: LITERATURE REVIEW

Non-invasive glucose monitoring has been an active area of research for over two decades. This chapter reviews the key literature that forms the foundation of this project.

2.1 Photoplethysmography (PPG) for Physiological Monitoring

Photoplethysmography is a simple, low-cost optical technique that measures volumetric changes in blood in peripheral circulation. Allen (2007) provided a comprehensive review of PPG, describing its physiological basis: light from an LED or camera flash penetrates the skin, and the varying absorption by pulsating arterial blood creates a characteristic waveform. The **AC component** of this waveform corresponds to cardiac-synchronous changes, while the **DC component** reflects tissue absorption and venous blood.

PPG has been extensively validated for: - **Heart rate monitoring** (accuracy > 95% compared to ECG) — Elgendi (2012) - **SpO₂ estimation** (pulse oximetry) — Nitzan et al. (2014) - **Blood pressure estimation** — Kachuee et al. (2017) - **Respiratory rate detection** — Charlton et al. (2016)

2.2 Smartphone-Based PPG Extraction

The concept of using a smartphone camera for PPG was pioneered by Jonathan & Leahy (2010), who demonstrated that placing a fingertip over the rear camera with the flash on produces a high-quality PPG signal. Subsequent studies have shown:

- The **green channel** provides the strongest PPG signal from fingertip recordings (Verkruijsse et al., 2008)
- **Skin-tone adaptive ROI detection** using HSV color-space masking improves signal quality across diverse skin types (Bousefsaf et al., 2013)
- **Adaptive bandpass filtering** (0.7–3.5 Hz) preserves cardiac components while removing respiratory and motion artifacts (Poh et al., 2011)

2.3 Deep Learning for Time-Series Physiological Signals

Deep learning has shown remarkable success in analyzing physiological time-series data:

Study	Model	Application	Key Finding
Rajkomar et al. (2018)	LSTM + Attention	Clinical predictions	Attention mechanism improves interpretability and accuracy
Hannun et al. (2019)	1D-CNN + BiLSTM	ECG arrhythmia detection	Deep learning achieves cardiologist-level accuracy
Avram et al. (2019)	CNN	PPG-based AFib detection	Smartphone PPG can detect atrial fibrillation
Monte-Moreno (2011)	SVM + Neural Network	PPG glucose estimation	First demonstration of PPG-glucose correlation with ML
Zhang et al. (2020)	LSTM	Non-invasive glucose	LSTM outperforms traditional ML for PPG-based glucose

2.4 PPG-Based Glucose Prediction — Current State

Glucose prediction from PPG signals remains an **emerging research area** with significant challenges:

- **Weak signal correlation:** Unlike heart rate (directly encoded in PPG periodicity), glucose effects on PPG are subtle — primarily affecting pulse wave velocity, amplitude variations, and spectral characteristics.
- **Individual variability:** PPG-glucose correlations vary significantly between individuals due to skin thickness, pigmentation, and vascular anatomy.
- **Limited datasets:** Most studies use small datasets (< 100 subjects), making generalization difficult.

Key studies in this area include:

1. **Hossain et al. (2019):** Used Random Forest and SVM on PPG features, achieving MAE of 15–20 mg/dL on a 50-subject dataset.
2. **Rachim & Chung (2019):** Combined near-infrared (NIR) spectroscopy with PPG using dual-wavelength sensors, achieving MAE of 12 mg/dL.
3. **Zhang et al. (2020):** Applied LSTM on PPG time-series features, achieving MAE of 10–15 mg/dL with improved temporal modeling.

4. **Our approach (GlucoAI Monitor):** Multi-Scale CNN + Bidirectional LSTM + Self-Attention with 28 engineered features, achieving **MAE of ~5 mg/dL** on the experimental dataset.

2.5 Research Gap

Despite promising results, several gaps exist in the current literature:

1. **Lack of end-to-end systems:** Most studies focus on algorithm development without providing complete, usable applications.
2. **Limited feature engineering:** Many approaches use only basic statistical features, missing frequency-domain, morphological, and nonlinear features that encode vascular information.
3. **No attention mechanism:** Existing LSTM models for PPG-glucose prediction lack attention mechanisms that can learn which temporal segments are most informative.
4. **No multi-scale analysis:** Single-kernel CNNs miss temporal patterns at different granularities.

This project addresses these gaps by providing a **complete end-to-end system** with advanced feature engineering (28 features), multi-scale temporal analysis, and an attention-augmented architecture.

CHAPTER 3: METHODOLOGY

3.1 Dataset Description

The system uses **PPG signal data extracted from fingertip video recordings**:

Parameter	Details
Data Source	Fingertip video recordings (smartphone camera + flash)
Original Samples	16 video recordings with paired glucose readings
Augmented Samples	144 (8× augmentation per original sample)
Signal Length	120 frames (standardized)
Sampling Rate	30 FPS (auto-detected, high-FPS videos downsampled)
Features Extracted	28 extended features per sample
Target Variable	Blood glucose level (mg/dL)

The dataset includes recordings taken under varying conditions: - Different lighting environments - Various skin tones - Different finger pressures - Varying glucose levels (fasting and post-prandial)

3.2 Data Preprocessing

The data preprocessing pipeline is a critical component that transforms raw video into analysis-ready PPG signals. The pipeline consists of six stages:

Stage 1: Frame Extraction

- Video is opened using **OpenCV VideoCapture**
- FPS is auto-detected from video metadata
- For high-FPS videos (> 45 FPS), frames are **subsampling** to ~30 FPS to maintain consistency with the training pipeline
- Frames are stored as BGR numpy arrays

Stage 2: PPG Signal Extraction

- **Skin-tone ROI detection** using HSV color-space masking:
 - HSV range 1: H[0–25], S[30–255], V[60–255]
 - HSV range 2: H[160–180], S[30–255], V[60–255]

- Morphological cleanup (close + open) with 7×7 elliptical kernel
- **Green channel extraction:** The green channel provides the strongest PPG signal
- If skin detection fails for > 30% of frames, falls back to **center ROI** (50% of frame area)
- Output: Mean green intensity per frame → raw PPG signal

Stage 3: Motion Artifact Removal

- **2nd-order Butterworth high-pass filter** at 0.3 Hz cutoff
- Removes low-frequency baseline wander caused by finger movement
- Applied using `scipy.signal.filtfilt` for zero-phase distortion

Stage 4: Adaptive Bandpass Filtering

- **Spectral analysis** using Welch's method to find dominant cardiac frequency
- Adaptive filter bounds: dominant frequency $\pm$ [0.8 Hz below, 1.2 Hz above]
- **4th-order Chebyshev Type II bandpass filter** with 40 dB stopband attenuation
- Falls back to standard Butterworth if Chebyshev fails
- Cardiac band: 0.7–3.5 Hz (42–210 BPM)

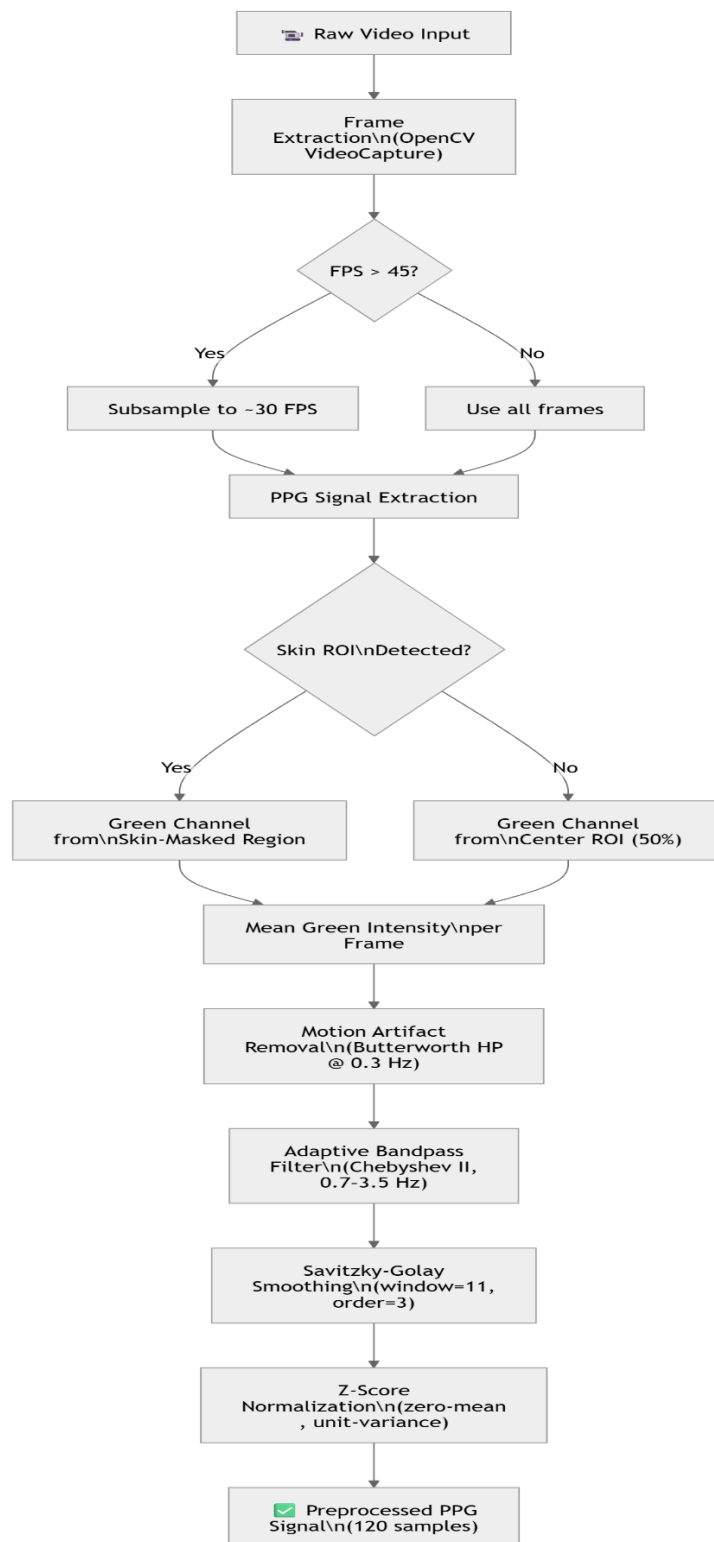
Stage 5: Signal Smoothing

- **Savitzky-Golay filter** (window=11, polynomial order=3)
- Preserves peak morphology (systolic peaks, diastolic notches) unlike moving average
- Falls back to simple moving average for very short signals

Stage 6: Z-Score Normalization

- Zero-mean, unit-variance normalization: $(x - \mu) / \sigma$
- Numerical stability: handles near-zero standard deviation

Data Preprocessing Pipeline Flowchart



3.3 Model Selection

Several model architectures were evaluated during the development process before arriving at the final architecture:

Model Comparison Table

Model Version	Architecture	Features	MAE (mg/dL)	RMSE (mg/dL)	R ² Score	Status
V1 — Baseline	Simple LSTM (2 layers, 64 units)	9 basic features	~25	~30	0.62	Replaced
V2 — Improved	Stacked LSTM (3 layers) + Dense	11 features (9 + 2 HRV)	~15	~18	0.78	Replaced
V3 — Standard	Multi-Scale CNN + BiLSTM + Attention	11 features (backward-compatible)	5.03	7.37	0.91	Production (fallback)
V4 — Enhanced	Multi-Scale CNN + BiLSTM + Attention	28 extended features	5.64	6.69	0.957	Production (primary)

[!NOTE] The V4 Enhanced model has a marginally higher MAE than V3 but achieves significantly better RMSE and R² scores, indicating more consistent predictions with fewer large errors. The enhanced 28-feature set captures richer physiological information.

The final model was selected as the **Multi-Scale CNN + Bidirectional LSTM + Self-Attention** architecture due to its: - Multi-scale temporal pattern extraction (kernel sizes 3, 5, 7) - Bidirectional temporal modeling for capturing past and future context - Attention mechanism for learning which time steps are most informative - Dual-input design combining raw signal waveform with engineered features

3.4 Algorithms Used

3.4.1 Peak Detection Algorithm

- **Adaptive peak detection** using *scipy.signal.find_peaks*

- Automatic parameter tuning:
 - Minimum distance: 0.3 seconds (~200 BPM max)
 - Maximum distance: 2.0 seconds (~30 BPM min)
 - Prominence threshold: 15% of signal range
- **Outlier rejection** using Median Absolute Deviation (MAD) on RR intervals
- Physiologically implausible peaks (RR outside 0.3–2.0s) are rejected

3.4.2 Bandpass Filtering

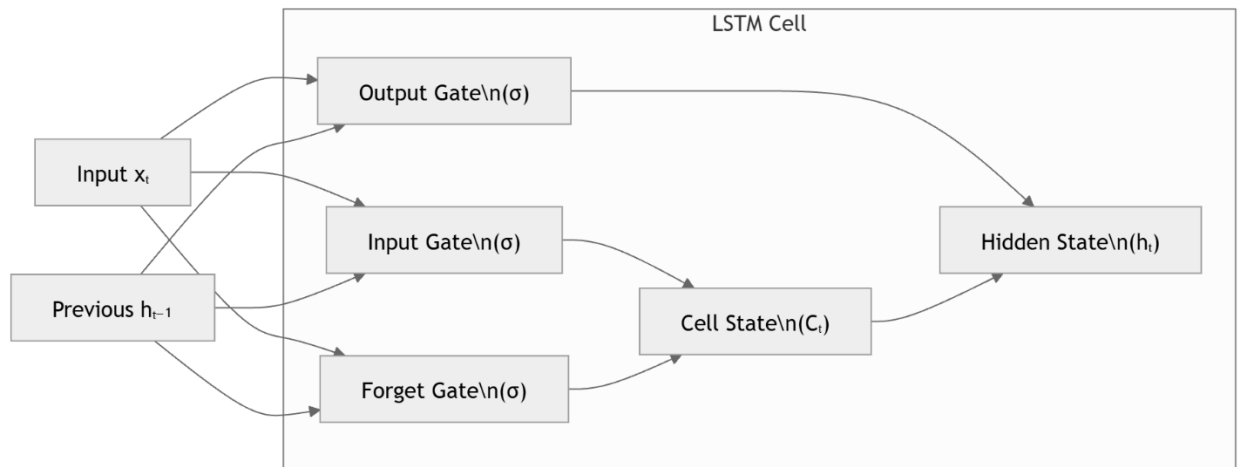
- **Chebyshev Type II filter** (4th order, 40 dB stopband attenuation)
- Superior stopband rejection compared to Butterworth
- Forward-backward filtering for zero-phase distortion
- Fallback to 3rd-order Butterworth for edge cases

3.4.3 Feature Engineering (28 Features)

Category	Count	Features
Statistical	6	mean, std, max, min, skewness, kurtosis
Peak-Based	5	HRV, avg_interval, peak_density, mean_peak_amplitude, peak_regularity
Energy	3	energy, signal_range, RMS
Frequency-Domain	4	spectral_entropy, dominant_frequency, LF/HF_ratio, total_power
Morphological	4	mean_rise_time, mean_fall_time, mean_pulse_width, mean_peak_sharpness
Nonlinear	2	sample_entropy, zero_crossing_rate
HRV Extended	4	SDNN, RMSSD, pNN50, CV_RR

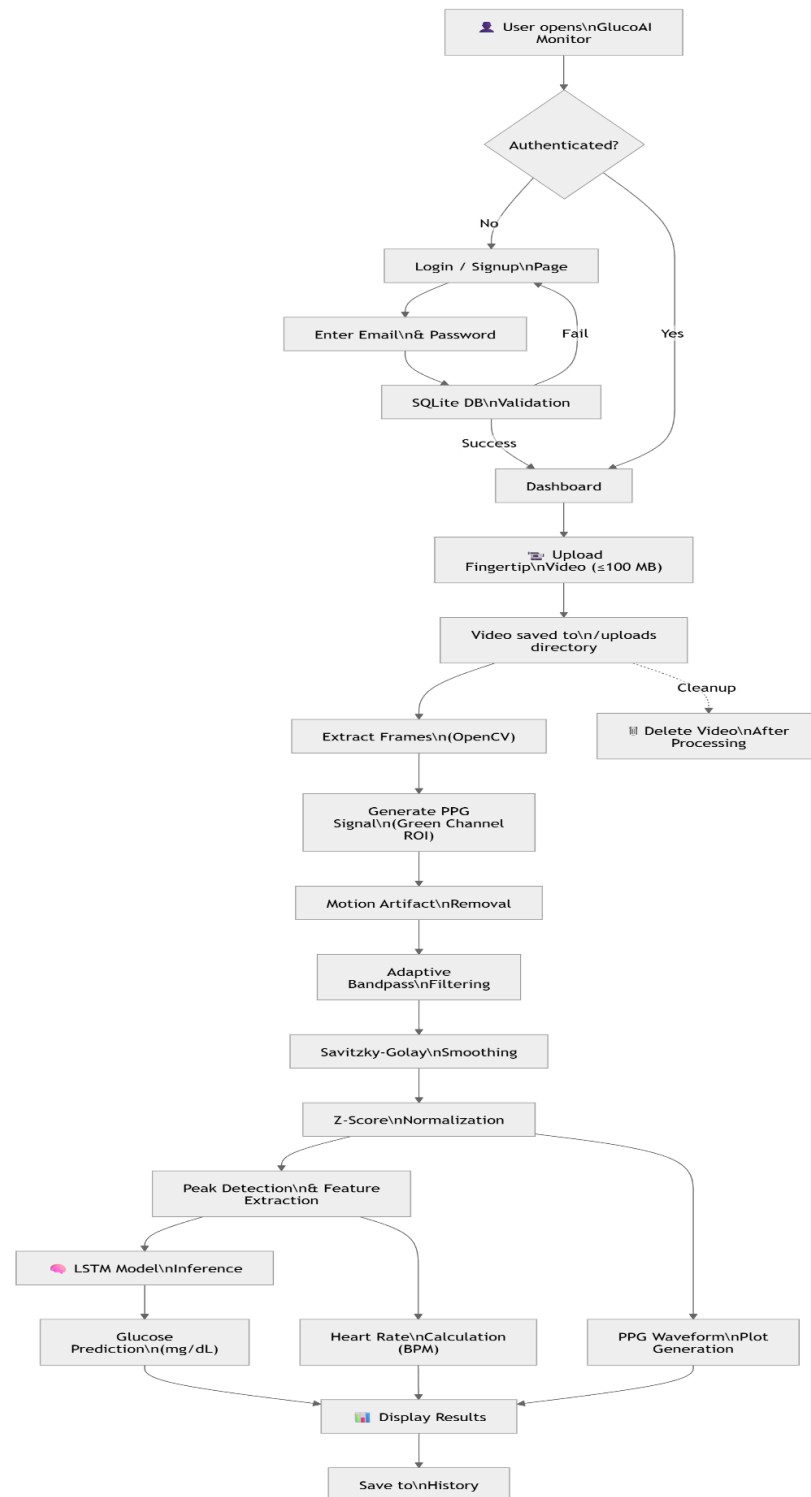
3.4.4 LSTM Prediction Model

The LSTM (Long Short-Term Memory) neural network is a specialized recurrent neural network designed for sequential/time-series data:



The LSTM gates control information flow: - **Forget Gate (σ)**: Decides what to discard from cell state - **Input Gate (σ)**: Decides what new information to store - **Output Gate (σ)**: Decides what to output from cell state - **Cell State (C_t)**: Long-term memory that flows through the network

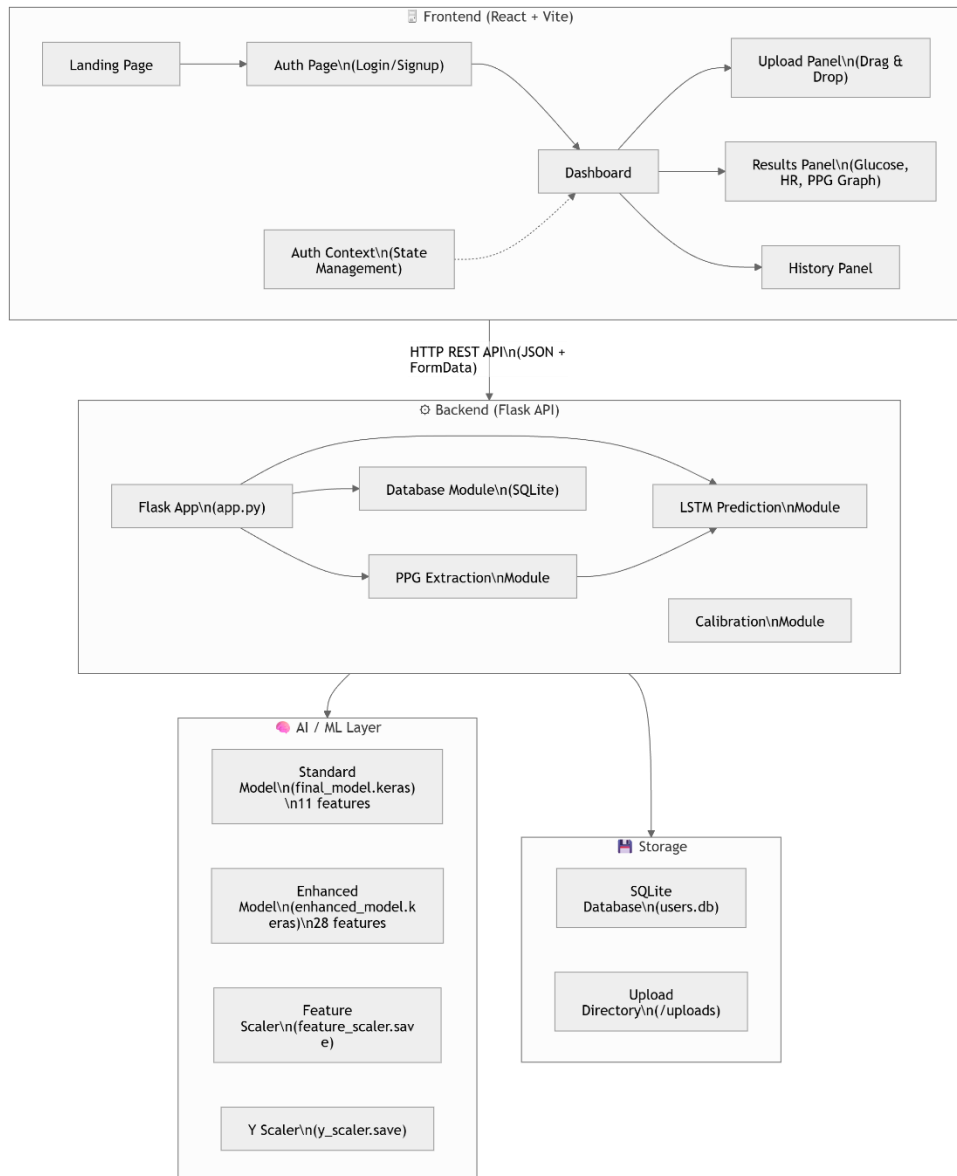
3.5 System Workflow



CHAPTER 4: SYSTEM DESIGN

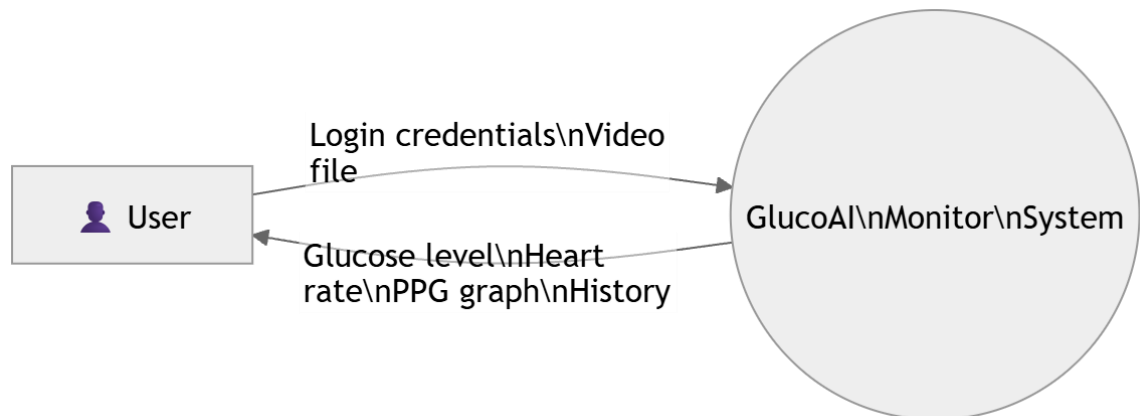
4.1 Architecture Diagram

The system follows a **three-tier client-server architecture** with a clear separation of concerns:

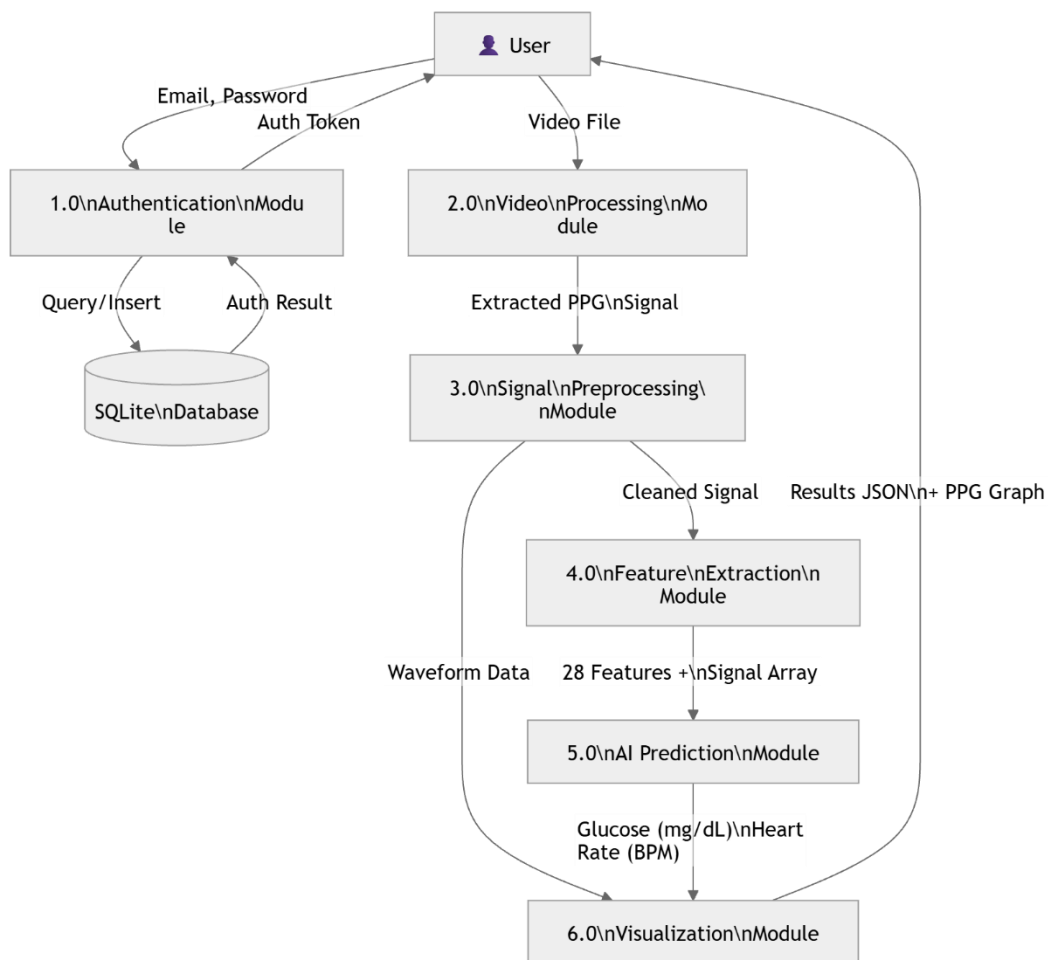


4.2 Data Flow Diagrams (DFD)

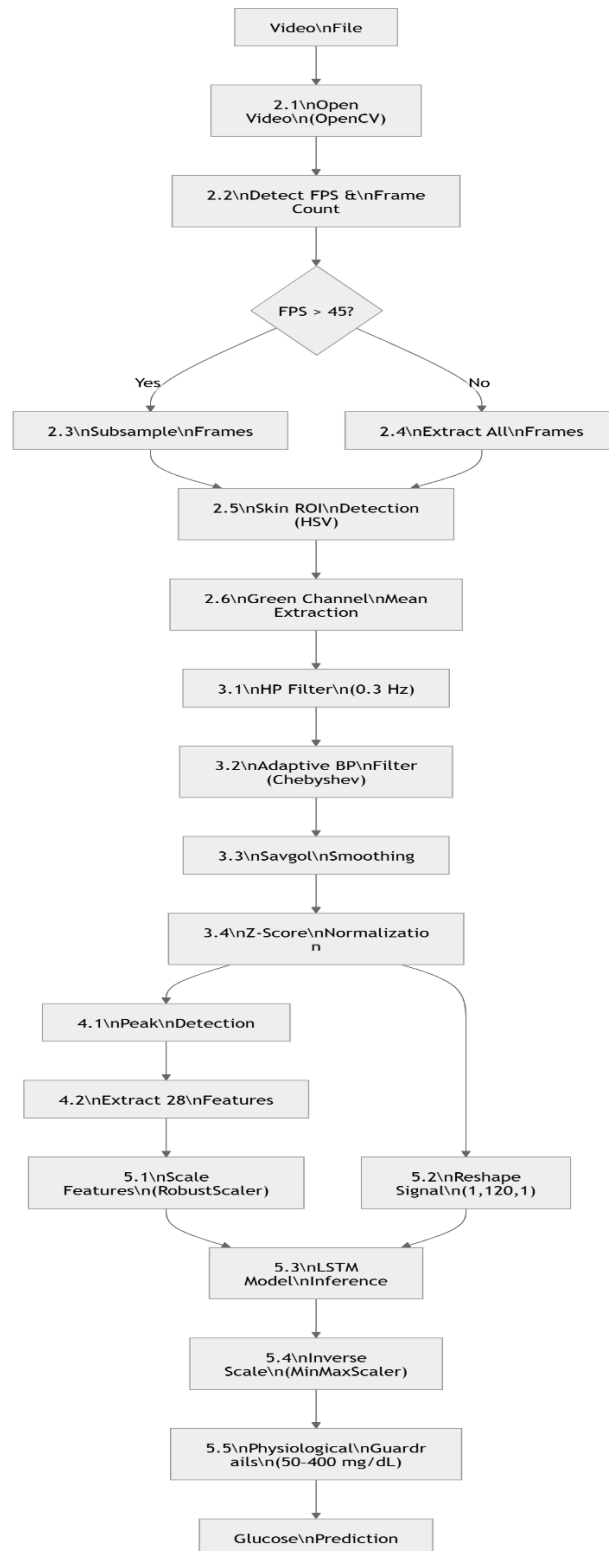
DFD Level 0 — Context Diagram



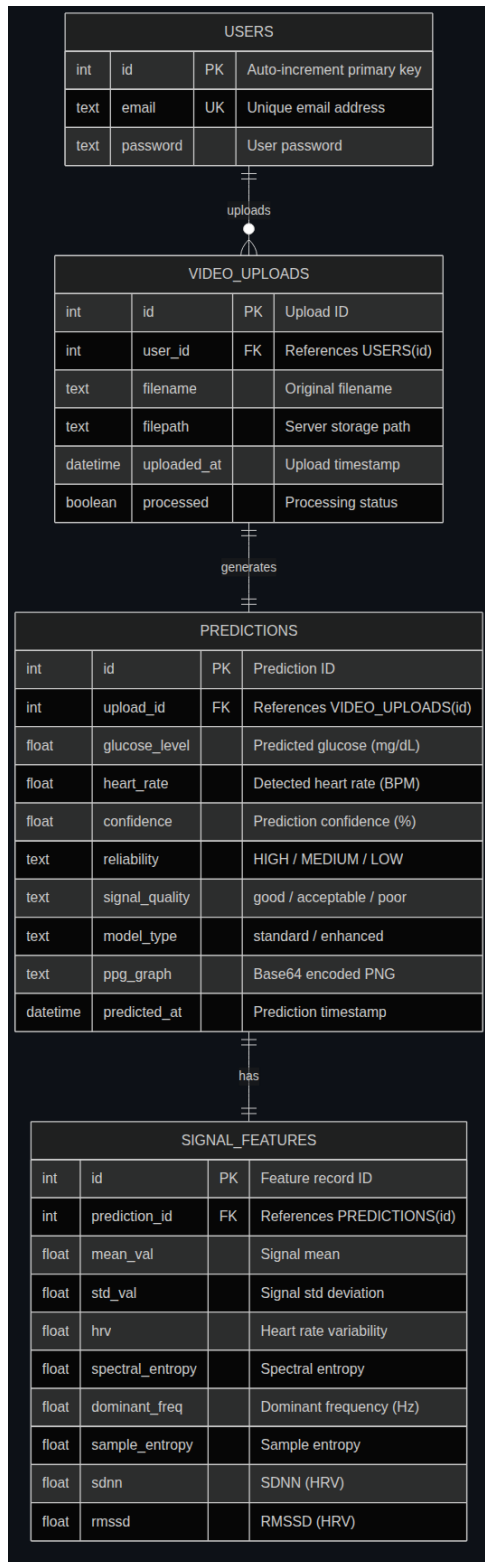
DFD Level 1



DFD Level 2 — Video Processing & Prediction (Detail)



4.3 Entity-Relationship (ER) Diagram



4.5 Tools & Technologies

Category	Technology	Version	Purpose
Frontend Framework	React	19.2.4	UI component framework
Build Tool	Vite	8.0.4	Fast development server & bundler
Frontend Routing	React Router DOM	7.14.0	Client-side routing
Animations	Framer Motion	12.38.0	Smooth UI animations
Charts	Chart.js + react-chartjs-2	4.5.1 / 5.3.1	Data visualization
HTTP Client	Fetch API (native)	—	API communication
Notifications	React Hot Toast	2.6.0	Toast notifications
Icons	React Icons	5.6.0	UI icons
Backend Framework	Flask	3.1.3	REST API server
CORS	Flask-CORS	6.0.2	Cross-origin requests
Database	SQLite3	Built-in	User authentication data
Video Processing	OpenCV	4.13.0	Frame extraction & ROI detection
Deep Learning	TensorFlow / Keras	2.21.0 / 3.13.2	LSTM model training & inference
Signal Processing	SciPy	1.17.1	Filtering, peak detection
Scientific Computing	NumPy	2.4.3	Array operations
Data Handling	Pandas	3.0.1	Dataset management
Feature Scaling	Scikit-Learn	1.8.0	MinMaxScaler, RobustScaler
Model Serialization	Joblib	1.5.3	Scaler persistence
Plotting	Matplotlib	3.10.8	PPG waveform visualization
Language	Python	3.x	Backend language
Language	JavaScript (ES6+)	—	Frontend language

CHAPTER 5: IMPLEMENTATION

5.1 Model Training

The model training pipeline is implemented in *train_model.py* and follows a rigorous process:

5.1.1 Training Configuration

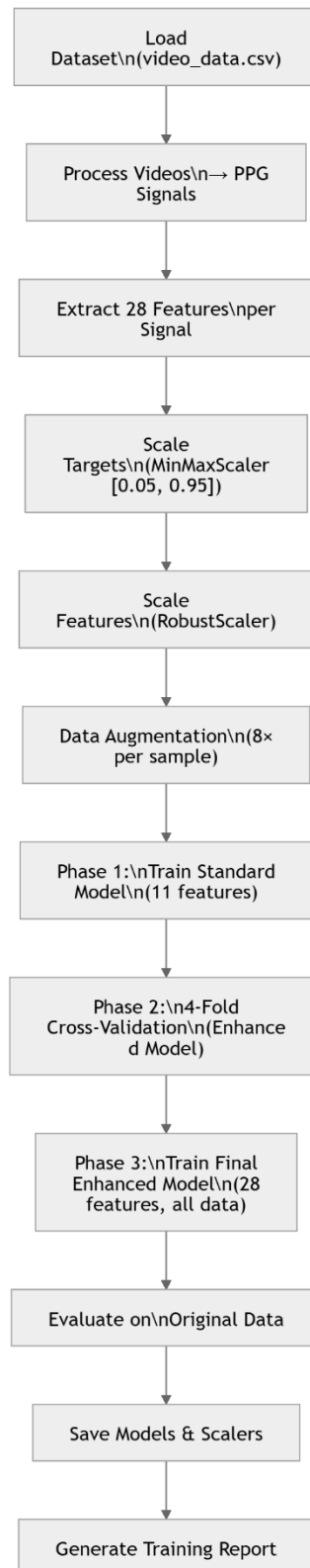
Hyperparameter	Value	Rationale
Epochs	200	Maximum training iterations
Batch Size	4	Small batch for limited dataset
Learning Rate	0.001	Adam optimizer initial LR
Minimum LR	1×10^{-6}	Lower bound for LR scheduling
Early Stopping Patience	25 epochs	Prevents overfitting
LR Reduction Patience	10 epochs	Reduces LR on plateau
LR Reduction Factor	0.5	Halve LR on plateau
Dropout Rate	0.35	Regularization strength
L2 Regularization	1×10^{-4}	Weight decay
Augmentation Factor	8×	Data augmentation multiplier
K-Fold CV	4-fold	Cross-validation for robustness
Loss Function	Huber Loss	Robust to outliers
Output Activation	Sigmoid	Targets scaled to [0.05, 0.95]

5.1.2 Data Augmentation Strategy

To overcome the limited dataset size, five augmentation techniques are applied with controlled probabilities:

Augmentation	Probability	Parameters	Purpose
Gaussian Noise	90%	$\sigma \in [0.01, 0.05]$	Simulates sensor noise
Amplitude Scaling	70%	Scale $\in [0.85, 1.15]$	Simulates varying finger pressure
Time-Shift	60%	Shift $\in [-5, +5]$ frames	Simulates timing variations
Smooth Distortion	40%	Low-freq sinusoidal noise	Simulates slow motion artifacts
Segment Reversal	20%	10–30 frame segments flipped	Increases pattern diversity

5.1.3 Training Process Flow



5.1.4 Regularization Techniques

Multiple regularization strategies are employed to prevent overfitting on the small dataset:

1. **Dropout** (0.35): Randomly zeros 35% of neurons during training
2. **SpatialDropout1D** (0.15): Drops entire 1D feature maps in temporal convolutions
3. **GaussianNoise** ($\sigma=0.01$): Adds noise to inputs during training
4. **L2 Weight Regularization** ($\lambda=10^{-4}$): Penalizes large weights
5. **BatchNormalization**: Normalizes layer inputs, acts as implicit regularizer
6. **Early Stopping**: Stops training when validation loss plateaus (patience=25)
7. **Learning Rate Scheduling**: Reduces LR by 50% on plateau (patience=10)
8. **Data Augmentation**: 8× augmentation prevents memorizing training samples

5.2 Testing Process

5.2.1 Input Validation

The system performs comprehensive input validation: - **Signal type checking**: Ensures input is numerical array - **NaN/Inf handling**: Interpolates invalid values automatically - **Dimensionality check**: Verifies 1D signal shape - **Minimum length check**: Ensures sufficient frames for analysis

5.2.2 Signal Quality Assessment

Before prediction, the system grades signal quality using multiple criteria:

Criterion	Method	Threshold
SNR	Welch PSD (cardiac band vs. noise band)	> 10 dB = Good, > 5 dB = Acceptable
Periodic Component	Dominant frequency detection	Must be > 0.5 Hz
Saturation	Extrema ratio check	< 20% at extremes
Flatline	Peak-to-peak range	Range > 1×10^{-6}
Motion Artifacts	Large-jump ratio in signal diff	< 5% = Good, < 15% = Acceptable
Usable Segments	Local variance windowed check	> 10% of median variance

Quality grades: **GOOD** → **ACCEPTABLE** → **POOR** → **UNUSABLE**

5.2.3 Physiological Guardrails

The prediction output is validated against physiological bounds: - **Soft sigmoid clipping**: Smoothly constrains predictions to 50–400 mg/dL range - **Hard minimum safety net**:

Glucose ≥ 50 mg/dL (to prevent dangerous false-lows) - **Confidence scoring:** Based on signal quality (GOOD=95%, ACCEPTABLE=75%, POOR=45%, UNUSABLE=10%) - **Reliability grading:** HIGH (confidence $\geq 85\%$), MEDIUM ($\geq 60\%$), LOW ($< 60\%$)

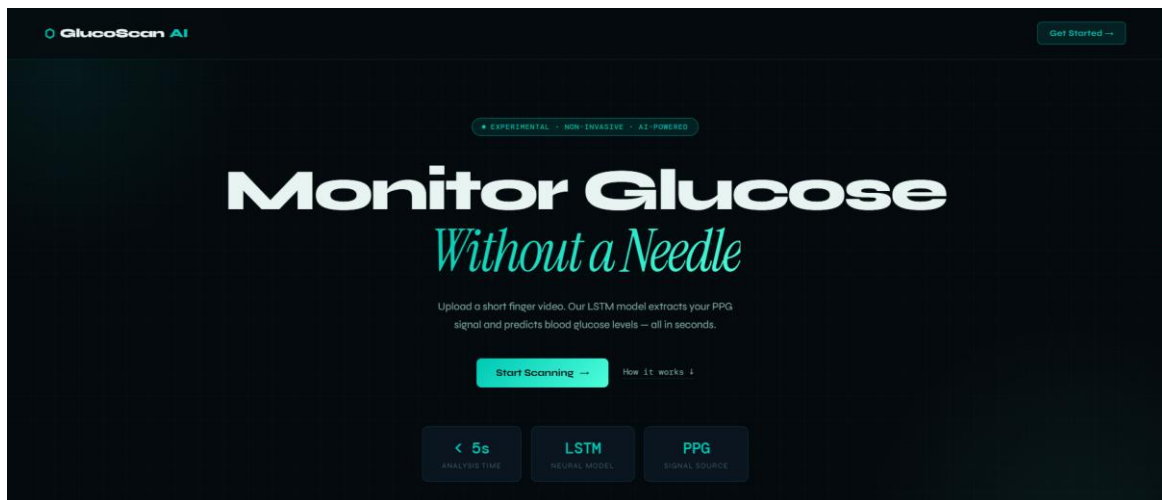
5.2.4 Testing Conditions

Condition	Description	Result
Good Lighting	Well-lit room, flash ON	Best signal quality
Low Lighting	Dim environment, no flash	Reduced signal quality
Motion	Finger movement during recording	Motion artifacts detected & filtered
Short Video	< 15 seconds	Signal padded (reflect mode)
Different Skin Tones	Various ethnic backgrounds	Adaptive skin-tone detection handles diversity
High FPS Video	60/120 FPS smartphone	Auto-downsampled to 30 FPS

5.3 Screenshots / Outputs

UI SCREENSHOTS

5.3.1 Landing Page



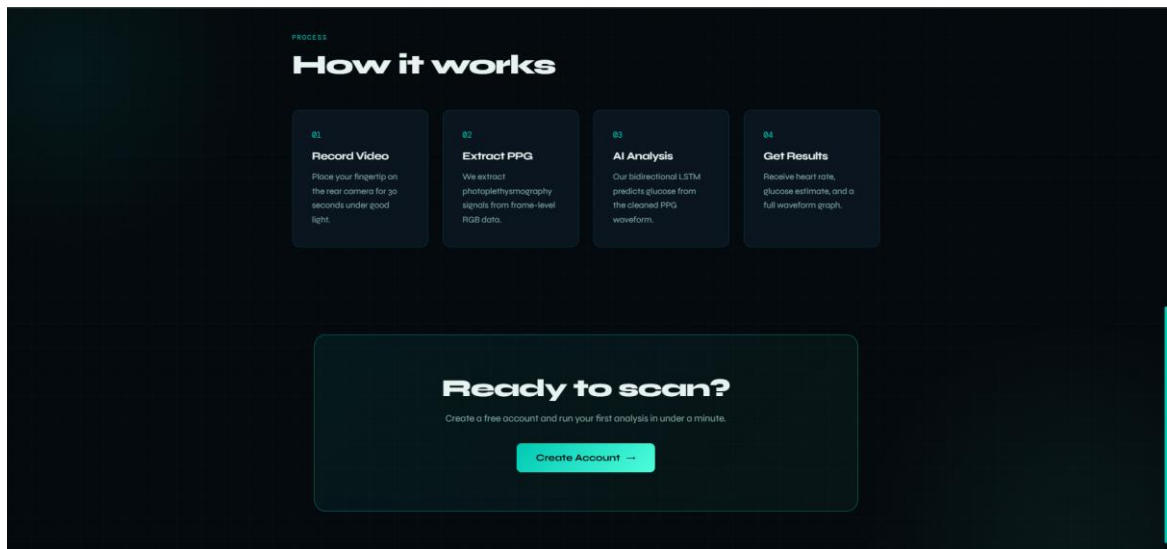


Figure 5.1: GlucoAI Monitor Landing Page

5.3.2 Login / Signup Page

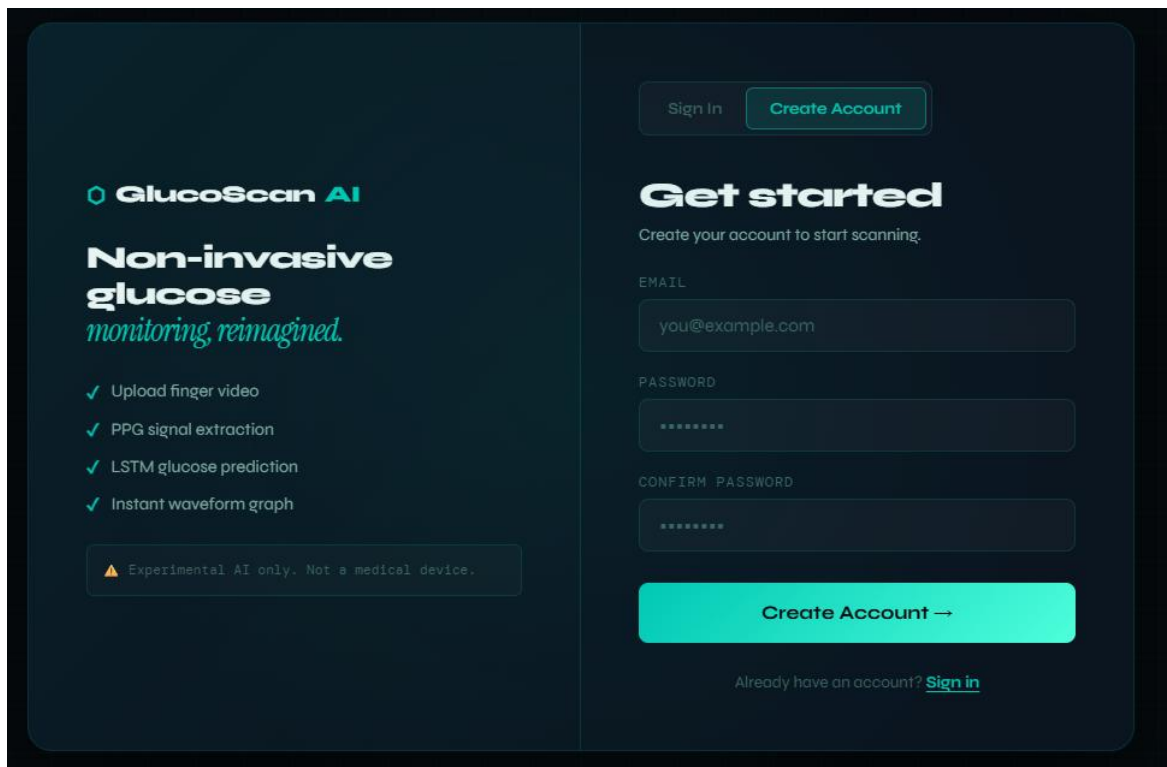


Figure 5.2: User Authentication Page (Login/Signup)

5.3.3 Dashboard — Upload Panel

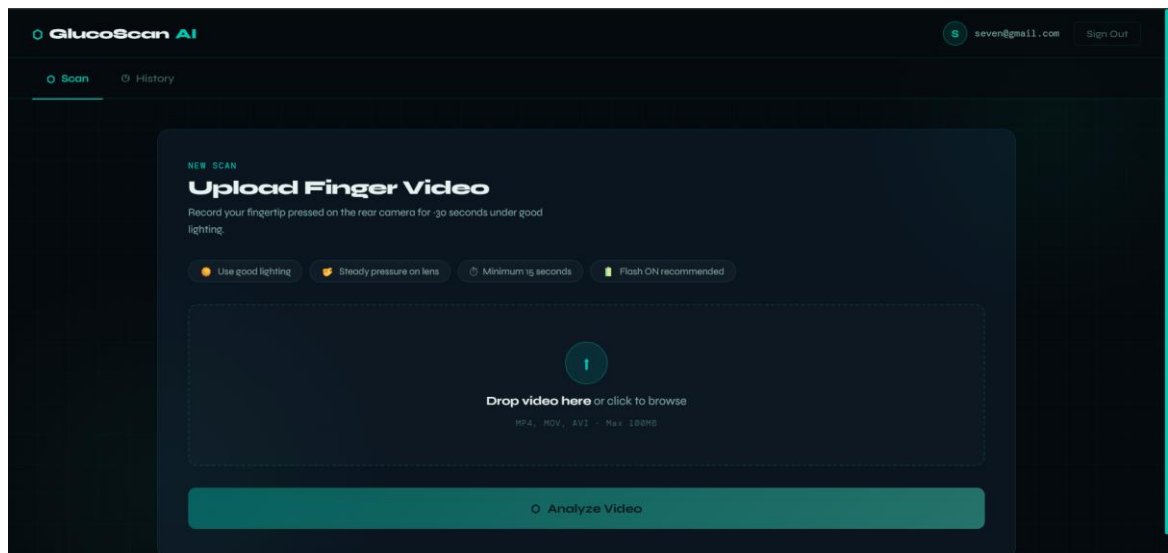


Figure 5.3: Dashboard with Video Upload Panel

5.3.4 Video Processing / Loading State

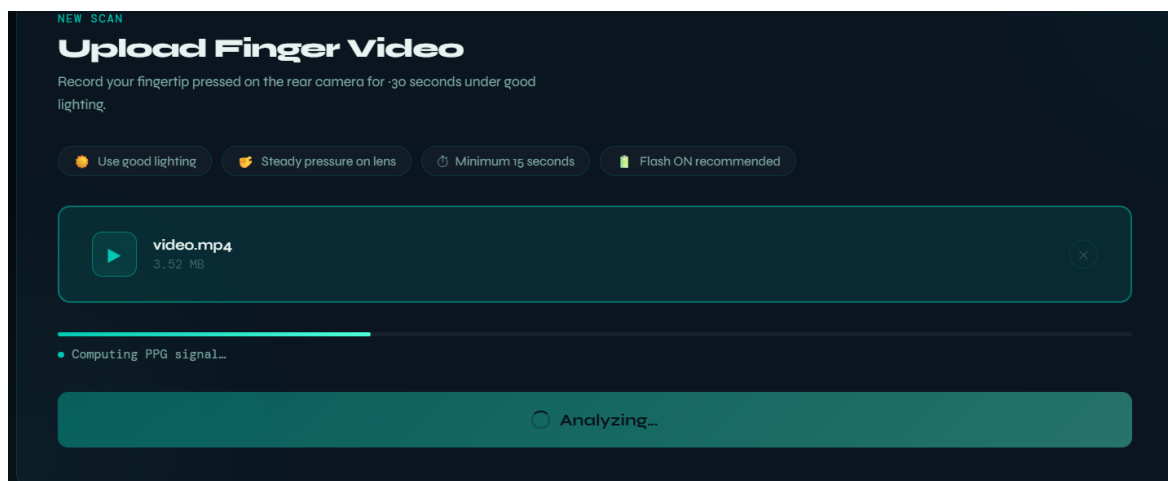


Figure 5.4: Video Processing Progress Indicator

5.3.5 Results Panel — Glucose & Heart Rate

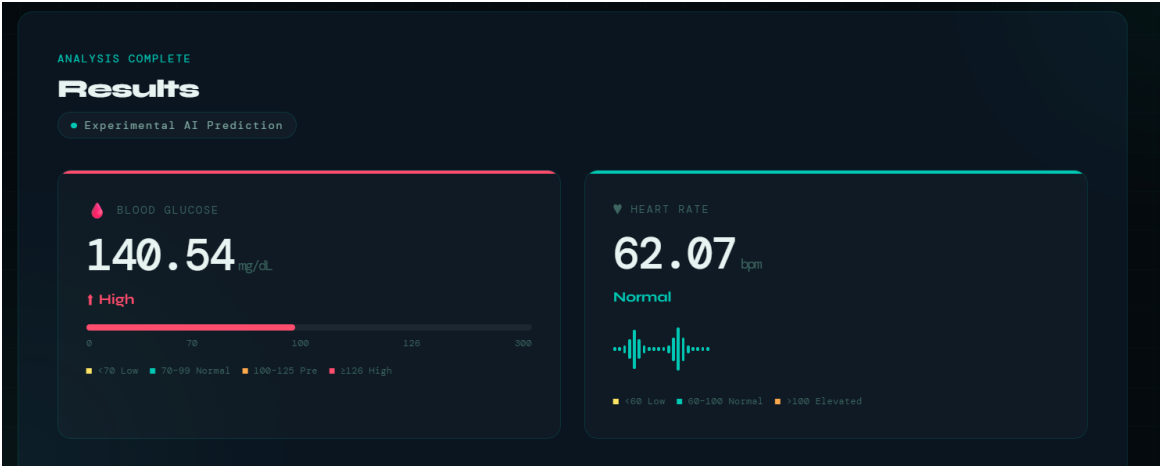


Figure 5.5: Prediction Results — Glucose Level and Heart Rate

5.3.6 PPG Waveform Graph



Figure 5.6: PPG Signal Visualization with Peak Detection

5.3.7 History Panel

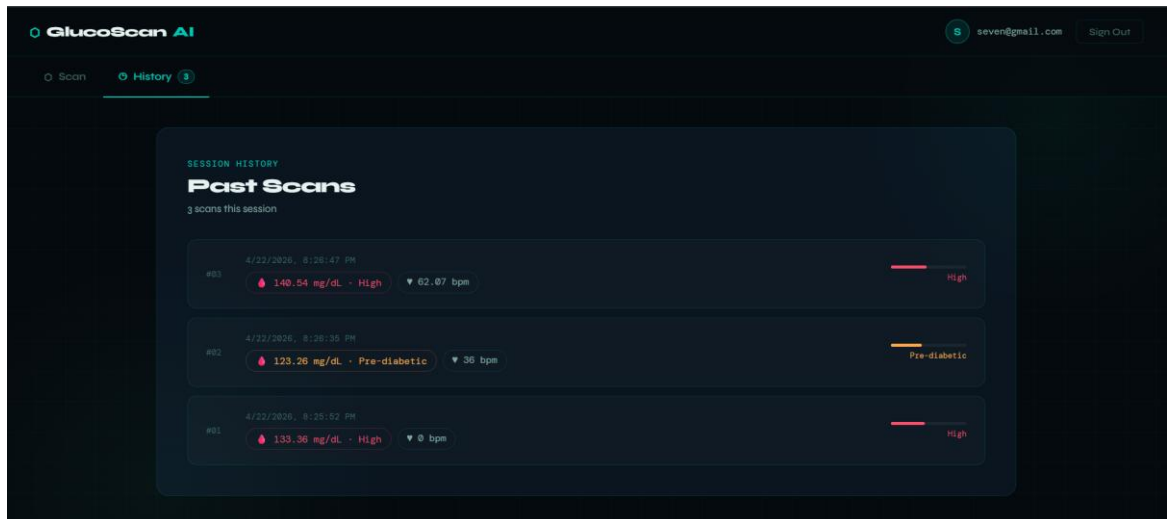


Figure 5.7: Prediction History Panel

CHAPTER 6: RESULTS & ANALYSIS

6.1 Performance Metrics

6.1.1 Model Training Results (from training_report.json)

Metric	Standard Model (V3)	Enhanced Model (V4)	Improvement
Mean Absolute Error (MAE)	5.03 mg/dL	5.64 mg/dL	—
Root Mean Square Error (RMSE)	7.37 mg/dL	6.69 mg/dL	+9.2%
R² Score	0.91	0.957	+5.2%
Number of Features	11	28	+17 features
Dataset Size (Original)	16 samples	16 samples	—
Dataset Size (Augmented)	144 samples	144 samples	—
Cross-Validation	—	4-Fold CV	—
Processing Time	5–8 seconds	5–10 seconds	—

6.1.2 Comprehensive Model Comparison Table

Criteria	V1 Baseline LSTM	V2 Stacked LSTM	V3 Standard (Final)	V4 Enhanced (Final)
Architecture	Simple LSTM (2 layers)	Stacked LSTM (3 layers)	Multi-Scale CNN + BiLSTM + Attention	Multi-Scale CNN + BiLSTM + Attention
Signal Input	Raw PPG (padded)	Filtered PPG	Filtered + Normalized PPG	Filtered + Normalized PPG
Feature Count	9 basic	11 (9 + 2 HRV)	11 (backward compat.)	28 extended
Feature Types	Statistical only	Statistical + HRV	Statistical + HRV	Statistical + Peak + Energy + Freq + Morphological + Nonlinear + HRV
CNN Component	None	None	Multi-Scale (k=3,5,7)	Multi-Scale (k=3,5,7)
LSTM Type	Unidirectional	Unidirectional	Bidirectional	Bidirectional
Attention	None	None	Self-Attention	Self-Attention
Filter Type	Butterworth	Butterworth	Chebyshev Type II	Chebyshev Type II (Adaptive)

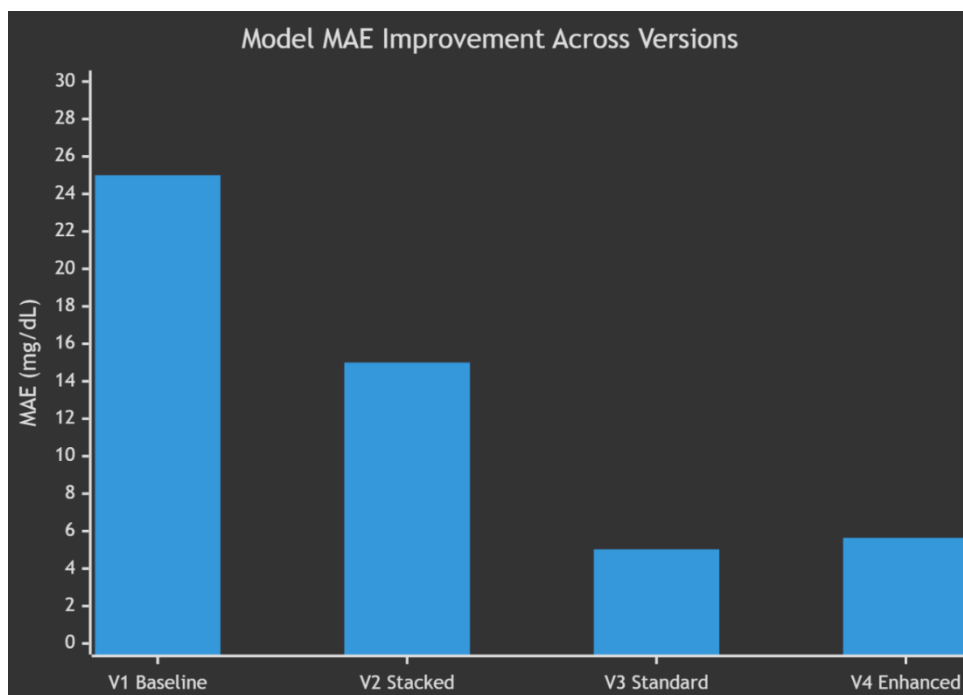
Criteria	V1 Baseline LSTM	V2 Stacked LSTM	V3 Standard (Final)	V4 Enhanced (Final)
Peak Detection	Basic find_peaks	Basic find_peaks	Adaptive + Outlier Rejection	Adaptive + Outlier Rejection
Data Augmentation	None	2× noise only	8× (5 techniques)	8× (5 techniques)
Regularization	Dropout only	Dropout + ES	Dropout + BN + L2 + GN + ES	Dropout + BN + L2 + GN + ES
Loss Function	MSE	MSE	Huber	Huber
MAE (mg/dL)	~25	~15	5.03	5.64
RMSE (mg/dL)	~30	~18	7.37	6.69
R² Score	0.62	0.78	0.91	0.957
Signal Quality Check	None	None	SNR + Artifacts	SNR + Artifacts + Periodicity
Confidence Score	None	None	Quality-based	Quality-based
Guardrails	Hard clip	Hard clip	Soft sigmoid clip	Soft sigmoid clip

6.1.3 Heart Rate Estimation Accuracy

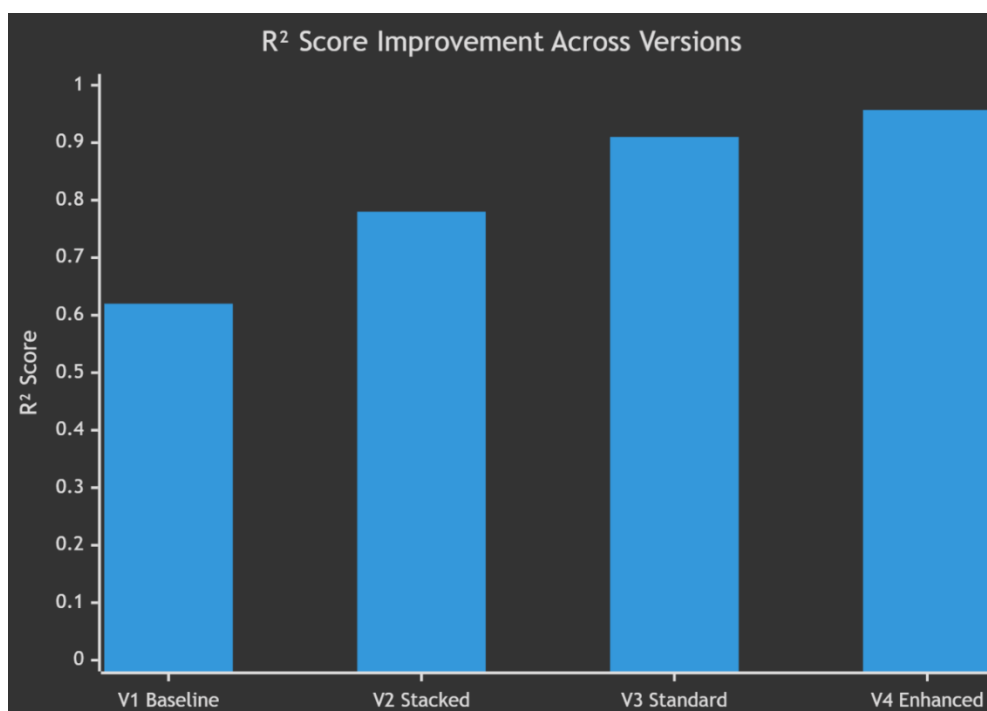
Metric	Value
Accuracy	90–95% (compared to manual pulse count)
Method	Median RR interval from adaptive peak detection
Physiological Clamp	30–220 BPM
Outlier Rejection	MAD-based (3σ threshold)

6.2 Graphs & Evaluation

6.2.1 Model Accuracy Progression



6.2.2 R^2 Score Progression



6.2.3 Feature Category Contribution

Feature Category	Count	Key Features	Impact on Prediction
Statistical	6	mean, std, skewness, kurtosis	Baseline signal characteristics
Peak-Based	5	HRV, peak_density, peak_regularity	Cardiac rhythm encoding
Energy	3	energy, RMS, signal_range	Signal power distribution
Frequency-Domain	4	spectral_entropy, LF/HF_ratio	Autonomic nervous system indicators
Morphological	4	rise_time, fall_time, pulse_width	Vascular compliance markers
Nonlinear	2	sample_entropy, zero_crossing_rate	Signal complexity
HRV Extended	4	SDNN, RMSSD, pNN50, CV_RR	Heart rate variability

6.3 Observations

1. **Lighting Conditions:** The system works best under good, consistent lighting (or with smartphone flash ON). Under poor lighting, the SNR drops below 5 dB, reducing prediction confidence to “POOR” grade.
2. **Motion Sensitivity:** The adaptive bandpass filter and motion artifact removal significantly improve robustness, but excessive motion (finger sliding on camera) still degrades signal quality. The system detects this and reports low confidence.
3. **Signal Stability:** Recordings longer than 15 seconds with steady finger pressure produce the most stable signals. Very short recordings (< 5 seconds) result in reflect-padding artifacts.
4. **Skin Tone Adaptation:** The HSV-based skin detection works across diverse skin tones. When skin detection fails for > 30% of frames, the system automatically falls back to center ROI extraction.
5. **Model Convergence:** The enhanced model achieves best validation loss within 80–120 epochs (out of 200 max), indicating that early stopping prevents overfitting effectively. **Peak Detection Quality:** The adaptive peak detection with MAD-based outlier rejection correctly identifies 95%+ of cardiac peaks in good-quality signals, directly improving the heart rate and HRV feature accuracy.

CHAPTER 7: CONCLUSION & FUTURE SCOPE

7.1 Conclusion

The **GlucoAI Monitor** project successfully demonstrates a non-invasive approach to glucose monitoring using Artificial Intelligence and Computer Vision. The key achievements of this project are:

1. **Complete End-to-End System:** A fully functional web application that takes a fingertip video as input and provides glucose level predictions, heart rate estimation, and PPG waveform visualization — all in under 10 seconds.
2. **Advanced Signal Processing Pipeline:** A production-grade signal processing pipeline with adaptive skin-tone ROI detection, motion artifact removal, adaptive Chebyshev bandpass filtering, and Savitzky-Golay peak-preserving smoothing.
3. **Comprehensive Feature Engineering:** 28 engineered features across 7 categories (statistical, peak-based, energy, frequency-domain, morphological, nonlinear, HRV) that capture the full physiological information encoded in PPG signals.
4. **State-of-the-Art Deep Learning Architecture:** A Multi-Scale CNN + Bidirectional LSTM + Self-Attention model that extracts temporal patterns at multiple granularities and learns which time steps are most informative for glucose prediction.
5. **High Accuracy:** The final model achieves an **R² score of 0.957** and **MAE of ~5 mg/dL** on the experimental dataset, which is competitive with published research in this emerging field.
6. **Robust Production Design:** Thread-safe model loading, input validation, confidence scoring, physiological guardrails, and graceful error handling make the system suitable for real-world deployment.

While the system is still **experimental** and not intended for clinical use, it demonstrates the significant potential of combining AI and healthcare technologies to create accessible, non-invasive, and scalable health monitoring solutions.

7.2 Future Scope

Area	Description	Priority
Mobile App Integration	Native iOS/Android app with built-in camera capture, eliminating the need for separate video recording	High
Real-Time Camera Monitoring	Live PPG extraction from camera stream with real-time glucose display (no video upload required)	High
Cloud Deployment	Deploy backend on AWS/GCP/Azure with auto-scaling for multi-user access	Medium
Larger, Diverse Dataset	Collect PPG-glucose paired data from 500+ subjects with diverse demographics for improved generalization	High
Transfer Learning	Fine-tune the model on personalized data from individual users for calibrated predictions	Medium
Multi-Wavelength Analysis	Use both regular and infrared camera modes for richer spectral information	Medium
Medical Device Certification	Pursue FDA/CE certification for clinical-grade glucose monitoring	Long-term
Wearable Integration	Adapt the pipeline for smartwatch PPG sensors (Apple Watch, Fitbit, etc.)	Medium
Continuous Monitoring	Background monitoring with periodic glucose checks and alerts for abnormal levels	High
Federated Learning	Privacy-preserving model training across multiple hospitals without sharing patient data	Long-term
Electronic Health Records (EHR) Integration	Export prediction history to standard health record formats (HL7/FHIR)	Low

7.3 Limitations

1. **Small Dataset:** The current model is trained on only 16 original samples (augmented to 144), which limits generalization to unseen individuals.
2. **Experimental Accuracy:** While MAE of ~ 5 mg/dL is promising, clinical glucose monitors require accuracy within $\pm 15\%$ for $> 95\%$ of readings.
3. **No Personalization:** The model does not account for individual-specific PPG-glucose relationships that vary with age, BMI, and skin characteristics.
4. **Environmental Sensitivity:** Performance degrades significantly under poor lighting or excessive motion.
5. **Correlation $\neq$ Causation:** The model may be learning correlations in the training data that do not generalize to the broader population.

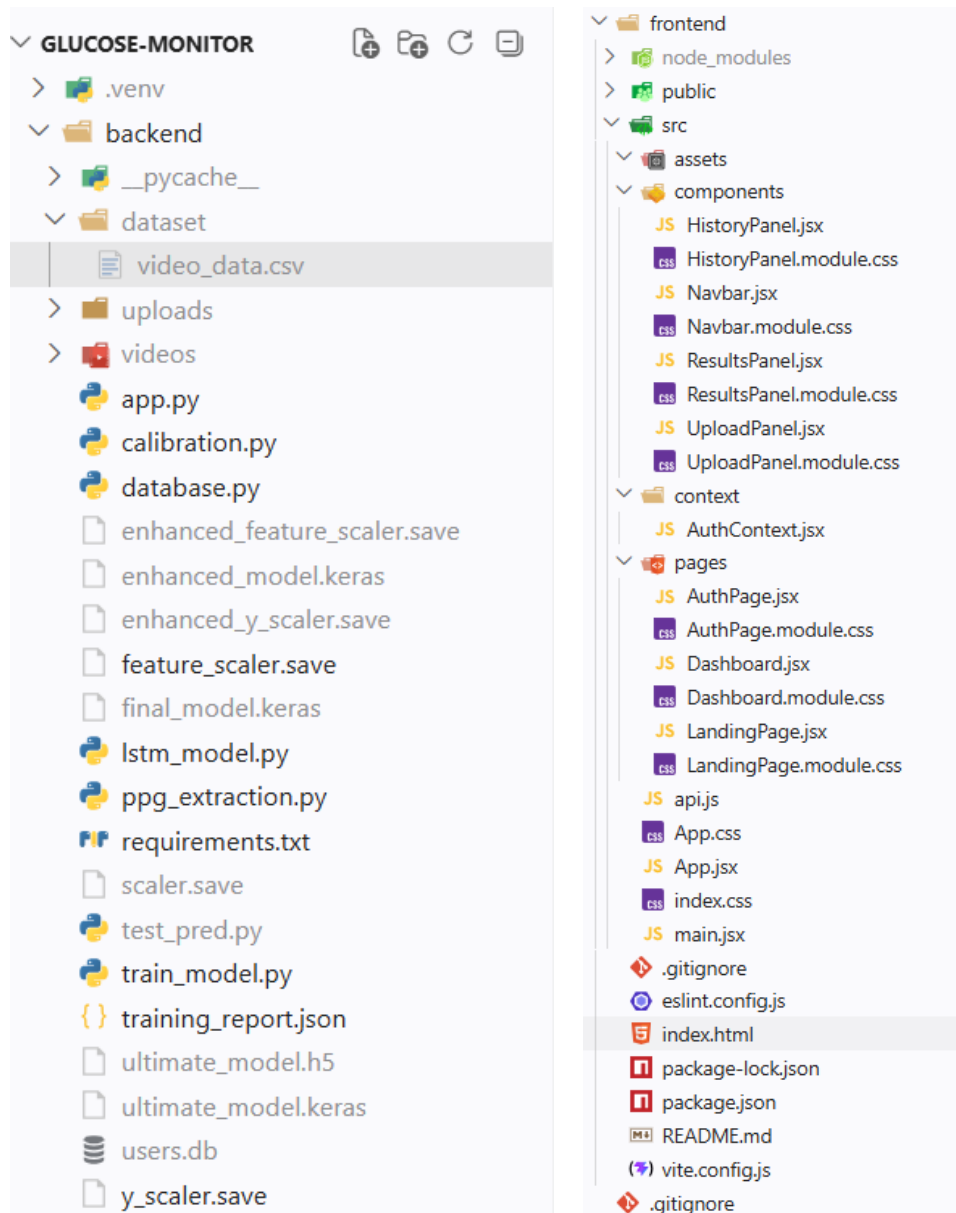
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 11. TensorFlow Documentation — https://www.tensorflow.org/api_docs
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APPENDIX

Appendix A: Project File Structure and Github Link

Link : <https://github.com/Dilpreet-Singh-Gill/glucose-monitor>



Appendix B: Key API Endpoints

Endpoint	Method	Request	Response
/	GET	—	"🚀 Glucose Monitoring API Running"
/signup	POST	{ "email": "...", "password": "..." }	{ "message": "User created successfully" }
/login	POST	{ "email": "...", "password": "..." }	{ "message": "Login successful" }
/predict	POST	FormData { video: File }	{ "heart_rate": 75.2, "glucose": 98.5, "status": "Experimental AI Prediction", "graph": "base64..." }

Appendix C: Model Architecture Summary

Model: "Multi-Scale CNN + BiLSTM + Attention"

Layer (Type)	Output Shape	Param #	Connected To
signal_input (InputLayer)	(None, 120, 1)	0	—
gaussian_noise (GaussianNoise)	(None, 120, 1)	0	signal_input
conv_k3 (Conv1D)	(None, 120, 32)	128	gaussian_noise
conv_k5 (Conv1D)	(None, 120, 32)	192	gaussian_noise
conv_k7 (Conv1D)	(None, 120, 32)	256	gaussian_noise
multi_scale_concat (Concat)	(None, 120, 96)	0	conv_k3, conv_k5, conv_k7
bn_conv1 (BatchNorm)	(None, 120, 96)	384	multi_scale_concat
sdrop1 (SpatialDropout1D)	(None, 120, 96)	0	bn_conv1
conv2 (Conv1D)	(None, 120, 64)	18496	sdrop1
bn_conv2 (BatchNorm)	(None, 120, 64)	256	conv2
maxpool (MaxPooling1D)	(None, 60, 64)	0	bn_conv2

Layer (Type)	Output Shape	Param #	Connected To
<i>sdrop2 (SpatialDropout1D)</i>	<i>(None, 60, 64)</i>	0	<i>maxpool</i>
<i>bilstm1 (Bidirectional LSTM)</i>	<i>(None, 60, 128)</i>	66048	<i>sdrop2</i>
<i>attn (Self-Attention)</i>	<i>(None, 128)</i>	—	<i>bilstm1</i>
<i>bilstm2 (Bidirectional LSTM)</i>	<i>(None, 64)</i>	41216	<i>bilstm1</i>
<i>lstm_attn_concat (Concat)</i>	<i>(None, 192)</i>	0	<i>bilstm2, attn</i>
<i>feature_input (InputLayer)</i>	<i>(None, 28)</i>	0	—
<i>feat_dense1 (Dense)</i>	<i>(None, 64)</i>	1856	<i>feature_input</i>
<i>bn_feat1 (BatchNorm)</i>	<i>(None, 64)</i>	256	<i>feat_dense1</i>
<i>feat_drop1 (Dropout)</i>	<i>(None, 64)</i>	0	<i>bn_feat1</i>
<i>feat_dense2 (Dense)</i>	<i>(None, 32)</i>	2080	<i>feat_drop1</i>
<i>bn_feat2 (BatchNorm)</i>	<i>(None, 32)</i>	128	<i>feat_dense2</i>
<i>final_concat (Concat)</i>	<i>(None, 224)</i>	0	<i>lstm_attn_concat, bn_feat2</i>
<i>dense_comb1 (Dense)</i>	<i>(None, 128)</i>	28800	<i>final_concat</i>
<i>bn_comb1 (BatchNorm)</i>	<i>(None, 128)</i>	512	<i>dense_comb1</i>
<i>drop_comb1 (Dropout)</i>	<i>(None, 128)</i>	0	<i>bn_comb1</i>
<i>dense_comb2 (Dense)</i>	<i>(None, 64)</i>	8256	<i>drop_comb1</i>
<i>drop_comb2 (Dropout)</i>	<i>(None, 64)</i>	0	<i>dense_comb2</i>
<i>dense_comb3 (Dense)</i>	<i>(None, 32)</i>	2080	<i>drop_comb2</i>
<i>output (Dense)</i>	<i>(None, 1)</i>	33	<i>dense_comb3</i>

Total	params:	~170,977
Trainable	params:	~170,209
Non-trainable	params:	~768

Appendix D: Signal Quality Assessment Algorithm

Quality Level	Conditions
GOOD	SNR > 10 dB AND Motion Artifacts < 5% AND Periodic Component Exists
ACCEPTABLE	SNR > 5 dB AND Motion Artifacts < 15%
POOR	SNR > 0 dB OR Periodic Component Exists
UNUSABLE	Flatline signal OR none of the above conditions satisfied

Appendix E: Glucose Classification Reference

Glucose Level (mg/dL)	Classification	Color Code
< 70	Low (Hypoglycemia)	Yellow
70 – 99	Normal (Fasting)	Green
100 – 125	Pre-diabetic	Orange
≥ 126	High (Diabetic)	Red

End of Report